BIOT 6900 · Module 7 Assignment (Computational Lab 7)

Building an Alzheimer's Disease Knowledge Graph from My Modules 2–6

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1. Schema and evidence

Until this module, I worked on each analysis separately in different files. For example, I had the ranked gene list from Module 2, druggability scores from Module 3, protein structures from Module 4, and risk scores from Module 6. In this assignment, I brought all these results together into one graph using Neo4j. My graph contains seven types of nodes (Disease, Gene, Phenotype, Pathway, Drug, Variant, and Patient) and six types of relationships. Overall, the graph has 83 nodes, including 30 patients, and 67 relationships.

Instead of using a general list of Alzheimer's disease (AD) genes, I selected the top 15 genes from my own Module 2 results: SAMHD1, PDZD2, ZMYM2, APOE, ASAP1, CACNG8, COL11A1, MAPT, APOC1, RHOJ, NDUFA10, WNK2, CLU, FARP2, and NUMA1. I also added APP and GRIN2B because they are targets of approved AD drugs and are present in my Module 2 data (ranked 19 and 358). I included SORL1 and FERMT2 because they have SNPs included in my Module 6 risk score. Finally, I added ACHE because it is the target of donepezil, even though it was not included in my Module 2 gene list.

Relationship	Direction	Evidence I stored
ASSOCIATED_WITH	Gene → Disease	GWAS p-value and a source label. For AD genes the source is Module 2 (GWAS + RNA + protein), and the RNA and protein fold changes sit on the gene node. ACHE is labelled 'pharmacology only'.
HAS_PHENOTYPE	Disease → Phenotype	HPO terms: dementia, memory impairment, cognitive impairment, cerebral atrophy and neurofibrillary tangles.
PARTICIPATES_IN	Gene → Pathway	Reactome pathway names. One grouping ('Endosomal sorting') is my own and is labelled that way.
TARGETS	Drug → Gene	Approval status (approved or investigational).
OCCURS_IN	Variant → Gene	How the SNP was mapped to a gene (MAPPED_GENE from my Module 2 GWAS file) and whether it is intergenic. Variant nodes also carry the Module 6 PRS weight and the Module 3 ESM-2 score for APOE ε4 and ε2.
CARRIES	Patient → Variant	How many copies of APOE ε4 (rs429358) or ε2 (rs7412) the patient carries.

Table 1. Relationship types and the evidence on each one.

I chose a property graph because it allows me to add evidence directly to each relationship. This makes it easier to identify the difference between a strong GWAS result and a weak connection. I also added information from my previous modules as gene properties. These include Ensembl and UniProt IDs from my Module 3 proteome file, small-molecule and antibody druggability scores from Module 3 (using the higher of the two scores), and AlphaFold confidence and pocket results for SAMHD1 and APOE from Module 4. I did not include Module 5 because my selected gene was CYP2C19, which is involved in drug metabolism and is not directly related to AD.

Patients: The patient data comes from Module 6. I selected 30 individuals from the 1000 Genomes dataset, with 6 people from each ancestry group. I calculated their APOE genotypes and PGS000334 risk scores using their genetic data. I followed the same allele-flipping method used in Module 6, and the average PRS for each ancestry group matched my earlier Module 6 results exactly. The patient IDs are pseudonyms, and their case/control status is simulated because the 1000 Genomes dataset does not provide disease information.

Related disease: I selected Aicardi–Goutières syndrome (AGS) as a related disease because mutations in SAMHD1, my top-ranked gene in Modules 2, 3, and 4, can cause this condition. AGS shares some features with AD, such as cognitive impairment and brain atrophy. Baricitinib, a JAK1/2 inhibitor, is used to treat AGS. I made two small changes to the provided build code. Without these changes, the SAMHD1→AGS relationship would have been labelled as "GWAS" and assigned SAMHD1's AD p-value, which would not be correct evidence.

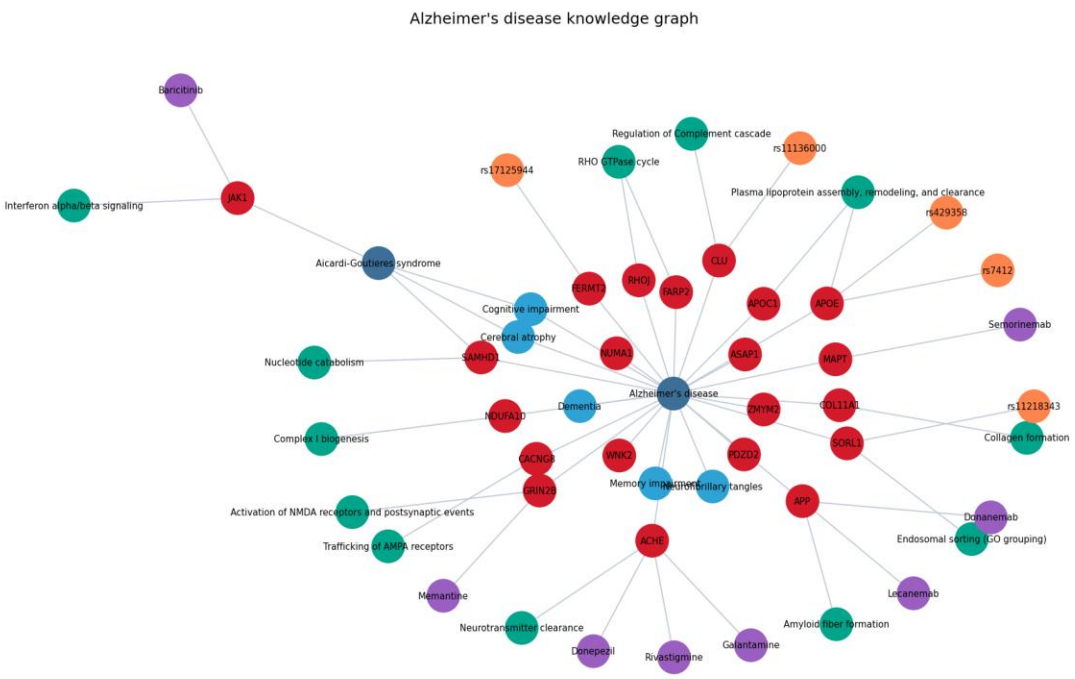


Figure 1. My AD knowledge graph (red = gene, purple = drug, green = pathway, light blue = phenotype, orange = variant, dark blue = disease). Patients are not drawn.

2. Top targets by centrality

Gene	PageRank (directed)	PageRank (undirected)	Degree	Why it ranks there
APOE	0.065 (1st)	0.025 (3rd)	4	Two PRS variants carried by 11 patients
ACHE	0.022 (2nd)	0.036 (1st)	5	Three approved drugs point to it
APP	0.016	0.029	4	Two antibodies + amyloid pathway
CLU, GRIN2B, SORL1, JAK1	0.011	0.022	3	One drug or variant plus a pathway
SAMHD1	0.006 (joint last)	0.019 (8th)	3	All its edges point outward
PDZD2, ZMYM2, WNK2, NUMA1, ASAP1	0.006	0.007	1	No pathway, drug or variant

Table 2. Gene centrality from §5 and §5b of my notebook.

APOE ranks first, which is what I expected because APOE ε4 is the strongest common genetic risk factor for late-onset AD. APP and CLU are also ranked near the top. APP produces amyloid-β and is targeted by the drugs lecanemab and donanemab. CLU is also a well-known AD GWAS gene. However, the ranking is also affected by how I created the graph. APOE ranks first in directed PageRank because 11 patients are connected to its two variants, which are then connected to APOE. In the undirected graph, APOE drops to third place. ACHE ranks second in directed PageRank and first in the undirected graph, even though it is not included in my Module 2 data. Its high ranking is mainly because three symptomatic drugs target this gene.

The most surprising result was SAMHD1. It ranked first in Module 2, had the highest small-molecule druggability score in Module 3 (0.824), and had a confirmed pocket in Module 4. However, directed PageRank places it joint last because all its connections point outward towards AD, AGS, and its pathway. PageRank mainly gives importance to nodes that receive incoming connections. JAK1 ranks higher than SAMHD1, even though JAK1 is not an AD gene. Five of my top 15 genes are also at the bottom because they have only one connection and no pathway or drug information. This shows that a low ranking is due to missing information in my graph rather than weak biological evidence. A better approach would be to give more weight to each connection based on its evidence, such as using −log10 p-values, instead of simply counting connections.

Communities and patients. Community detection separated APOE ε4 carriers (6 patients connected around rs429358) from ε2 carriers (5 patients connected around rs7412, APOE, and the lipoprotein pathway). In the stratification query (§6b), ε3/ε4 carriers had 4 cases out of 6, ε3/ε3 carriers had 4 out of 19, and ε2/ε3 carriers had 2 out of 5. Their average PRS percentiles were 82, 38, and 10, respectively. The PRS ranking follows the expected pattern, but the case rates are based on simulated disease status and only 30 people. Therefore, these results are only for illustration. The ancestry query also showed the same pattern as Module 6. The African-ancestry group had the highest average PRS, including PT12, who was an ε3/ε4 control at the 94th PRS percentile.

3. A hypothesis from a path

The 2-hop query identified seven drugs that target my AD-related genes. These include donepezil, rivastigmine, and galantamine, which target ACHE; memantine, which targets GRIN2B; and lecanemab and donanemab, which target APP. All these drugs are approved. It also identified semorinemab, an investigational anti-tau antibody that targets MAPT. The 5-hop drug repurposing query returned one potential candidate:

Repurposing candidate	Gene	Via disease	Shared phenotypes
Baricitinib	JAK1	Aicardi–Goutières syndrome	Cognitive impairment, Cerebral atrophy

Table 3. Result of the 5-hop query.

My hypothesis: In some AD patients with low SAMHD1 levels, type I interferon signalling may become overactive. A JAK1/2 inhibitor such as baricitinib may help reduce this activity. The connection in the graph only shows that the two diseases share some symptoms. However, the community analysis makes this idea more interesting because SAMHD1, AGS, JAK1, baricitinib, interferon α/β signalling, nucleotide catabolism, cognitive impairment, and cerebral atrophy were all grouped into the same community. In my Module 2 results, SAMHD1 was the most down-regulated gene in the AD hippocampus (RNA log2FC −3.65). In AGS, loss of SAMHD1 causes nucleotides to build up, which activates interferon signalling. This also connects with my Module 4 report, where I mentioned that SAMHD1 may need an activator instead of an inhibitor, which is difficult to develop. Baricitinib works further downstream by blocking the interferon response, so it does not need to target SAMHD1 directly.

Independent support: After finding this connection in my graph, I checked whether this idea had already been tested in any study. Baricitinib has been studied in AD through the NADALS basket trial (NCT05189106) at Massachusetts General Hospital. It was an open-label Phase 1/2 study involving people with subjective cognitive decline, MCI, AD, or ALS. The study enrolled 17 participants and examined whether baricitinib could reach the cerebrospinal fluid and reduce interferon-related biomarkers. The trial focused on general neuroinflammation rather than SAMHD1 specifically. Therefore, my graph adds a possible way to identify which patients may benefit from the treatment instead of suggesting a completely new drug.

How I would test it. First, I would use a larger AD RNA-seq dataset, such as ROSMAP, to check whether low SAMHD1 levels are confirmed and whether they are linked to higher interferon-stimulated gene expression. Second, I would reduce SAMHD1 levels in iPSC-derived microglia and measure interferon-stimulated genes and microglial activation. I would then test whether baricitinib can reduce these effects. Third, in a clinical trial similar to NADALS, I would divide patients based on their CSF interferon signature and check whether patients with higher interferon levels respond better to the treatment. One important challenge is that JAK1/2 inhibitors do not easily cross the blood-brain barrier, which may reduce their effectiveness in the brain.

4. Limitations

Weak edges. The drug repurposing path depends on the strength of its connections, and some of these connections are not very strong. The two shared symptoms are quite general. AGS mainly affects young children, whereas AD usually develops later in life. So, cerebral atrophy and cognitive impairment are common features of both diseases, but the reasons behind them are different. The JAK1→AGS connection is based on treatment evidence rather than genetic evidence. If I included only genetic ASSOCIATED_WITH relationships, as done in Step 8 of the example notebook, the baricitinib hypothesis would not appear. ACHE is also included in my graph only because of its connection to existing drugs. My Module 2 results combine three different cohorts that do not share any patients, and SAMHD1's high ranking mainly depends on RNA data from just 23 samples.

Mapping variants to genes. When I mapped the 22 PGS000334 SNPs using my GWAS file, I noticed the problems with using the nearest-gene method. Only 9 SNPs were mapped to genes in my Module 2 list, and 4 of them were intergenic (BIN1, ADAM10, CNN2/ABCA7, and PSMC3), so I excluded them. The rs3851179 SNP, which is commonly known as the PICALM SNP, was mapped to a pseudogene and a lncRNA, so it was completely missed.

The rs3795065 SNP is located between CNN2 and ABCA7, and my method would select CNN2 even though ABCA7 is the more commonly accepted candidate gene. The strong GWAS signal for APOC1 is also mainly due to its linkage with APOE. The ESM-2 scores for APOE variants show how likely a change is during evolution, not its actual disease risk. The ε4 variant scored +7.21, making it the most tolerated change in the entire protein. Also, the PRS weights are based on European data, and my Module 6 results showed that they are less accurate for other ancestry groups.

Annotation. I assigned the pathways manually, so genes without pathway information may appear less important than they actually are. APP, GRIN2B, and ACHE do not have Module 3 druggability scores because they were not included in my Module 3 top 15 genes. Importing data directly from Open Targets or Hetionet into Neo4j and giving each relationship a weight based on its evidence would help reduce my own selection bias. I think this is the biggest hidden limitation of creating a small graph manually.

Use of AI. I built this graph from my own outputs from Modules 2, 3, 4 and 6. I set up and ran the Neo4j database, checked every result, and wrote this report in my own words. I used an AI assistant (Claude) for support: it helped write the data-preparation script and the data cell of the notebook, suggested the AGS/baricitinib path to explore, and gave me a starting outline for the report. I verified its work myself. The gene IDs and SNP mappings come from my own Module 2 and 3 files, the trial and references were checked on ClinicalTrials.gov and PubMed, and the results in Tables 2 and 3 match my notebook output.

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